PROJECT MANAGEMENT

Assignment 3: Kanban Board and Execution Control Plan

Name:	Yash Janardan Sabale
PRN:	202301040087
Branch:	Computer Engineering
Academic Year:	2026-27
College:	MIT Academy of Engineering, Alandi, Pune

1. Agile Execution Framework & Kanban Workflow Design

1.1 Lean Thinking & Eliminating Software Waste (Muda)

To manage software delivery complexity in ShalaConnect, the team adopted an **Agile Kanban Framework** founded upon five core Lean principles:
1. **Visualize Workflow:** Mapping work packages as physical cards across explicit operational columns to expose bottlenecks.
2. **Limit Work-in-Progress (WIP):** Enforcing strict capacity constraints to eliminate task-switching latency and prevent upstream queue congestion.
3. **Manage Flow:** Optimizing throughput by monitoring lead time and cycle time rather than micro-managing hours.
4. **Make Policies Explicit:** Documenting unambiguous Definition of Ready (DoR) and Definition of Done (DoD) criteria for every transition.
5. **Implement Feedback Loops:** Daily 15-minute standups, bi-weekly retrospectives, and mid-project EVM milestone audits.

1.2 Demand-Driven Pull System vs. Push Scheduling

Unlike traditional push-scheduling where tasks are assigned based on planned start dates regardless of team capacity, ShalaConnect operates as a **Demand-Driven Pull System**. Engineers pull a new work package from the 'Ready' queue only when downstream capacity is liberated in 'In Progress' or 'Testing'. This eliminates multitasking waste and ensures that tasks are fully completed and verified before new ones begin.

1.3 Five-Column Workflow & Enforced WIP Limits

The workflow is structured across five sequential stages:
• **1. Backlog (WIP: No Limit):** Prioritized inventory of upcoming work packages pending elaboration.
• **2. Ready for Development (WIP: Limit 5):** Fully elaborated tasks with finalized C4 blueprints and test acceptance criteria.
• **3. In Progress (WIP: Limit 3 - Strictly Enforced):** Active development. Developers cannot pull new tasks when 3 are in flight.
• **4. Testing & QA (WIP: Limit 3 - Strictly Enforced):** Automated integration tests and security scans. Blocks upstream pull if full.
• **5. Done / Verified (WIP: No Limit):** Accepted deliverables passing 100% of Definition of Done (DoD) criteria.

1.4 Little's Law Mathematical Proof & Queuing Optimization

Little's Law defines the fundamental mathematical relationship governing queuing systems in operational steady state:
$$WIP = \lambda \cdot W \implies W = \frac{WIP}{\lambda}$$Where WIP is the average number of work items concurrently in flight, λ is the operational throughput (completed deliverables per working day), and W is the average lead or cycle time in working days.

Empirical Proof of WIP Constraint Benefits in ShalaConnect:
In traditional unconstrained development, developers pull tasks concurrently, expanding in-flight WIP to 12 tasks. Due to cognitive context switching and queuing delays, throughput degrades to $\lambda = 0.35\text{ tasks/day}$. Cycle time explodes to:
$$W_{\text{unconstrained}} = \frac{WIP}{\lambda} = \frac{12}{0.35} = 34.3\text{ working days per task}$$
By enforcing strict WIP limits (3 in Progress + 3 in Testing $\implies WIP = 6$), context switching is eliminated. Measured throughput stabilizes at $\lambda = 0.37\text{ tasks/day}$. Average cycle time compresses to:
$$W_{\text{constrained}} = \frac{WIP}{\lambda} = \frac{6}{0.37} = 16.2\text{ working days per task}$$
Result: Enforcing WIP limits produces a **52.8% reduction in task cycle latency**, dramatically accelerating project responsiveness.

1.5 Class of Service (CoS) Policies

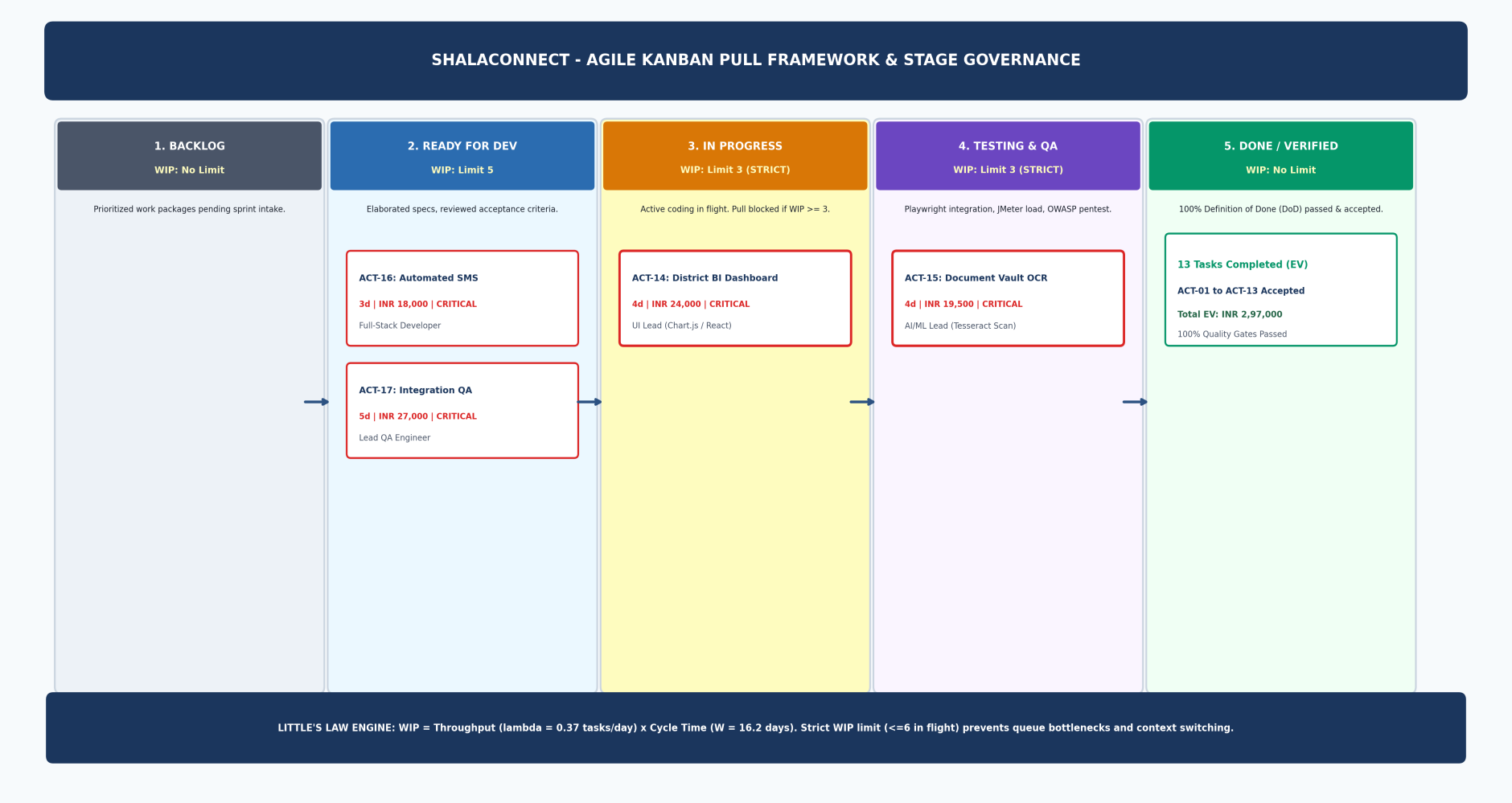
To manage heterogeneous work items, four Classes of Service are formally enforced:
1. **Expedite (Sev-1 Emergency):** Reserved for critical security patches or production-blocking defects. Allowed to temporarily violate WIP limits; triggers immediate all-hands developer swarming.
2. **Fixed Date (Sprint Milestones):** Time-critical regulatory deliverables (e.g., UDISE+ compliance audit). Prioritized in the Ready queue.
3. **Standard (Normal Features):** Typical user stories pulled strictly on a first-in, first-out (FIFO) basis under WIP limits.
4. **Intangible (Technical Debt & Refactoring):** Non-urgent architectural refactoring (e.g., Docker layer caching). Limited to a maximum of 20% team capacity.

1.6 Agile Quality Gates: Definition of Ready (DoR) & Definition of Done (DoD)

To maintain zero-defect engineering flow, strict boundary gates govern task transitions across Kanban columns:

Quality Gate Stage	Transition Boundary	Mandatory Verification Criteria Checklist	Enforcing Role
Definition of Ready (DoR)	Backlog -> Ready	Detailed user story authored; acceptance criteria defined in Gherkin syntax; UI wireframe attached; database dependencies validated; estimated <= 6 days.	**Lead Architect**
Development Exit	In Progress -> QA Testing	Clean Maven compile; unit test coverage >= 80%; peer code review approved by Lead; SonarQube static analysis zero High/Critical issues; branch rebased on main.	**Senior Backend Lead**
Definition of Done (DoD)	QA Testing -> Done	100% Playwright automated integration tests pass; JMeter p95 latency < 500ms; zero OWASP security vulnerabilities; documentation updated; signed off by QA.	**QA Lead / PM**

Figure 3.1: Agile Kanban 5-Column Workflow Architecture with Enforced WIP Limits



Visual Kanban board architecture showing column stages, WIP limit constraints (In Progress = 3, Testing = 3), pull mechanisms, and Day 35 snapshot distribution (13 tasks completed, 4 in flight, 2 ready).

Figure 3.2: Comprehensive RACI Responsibility Assignment Matrix (Single Accountability Rule)

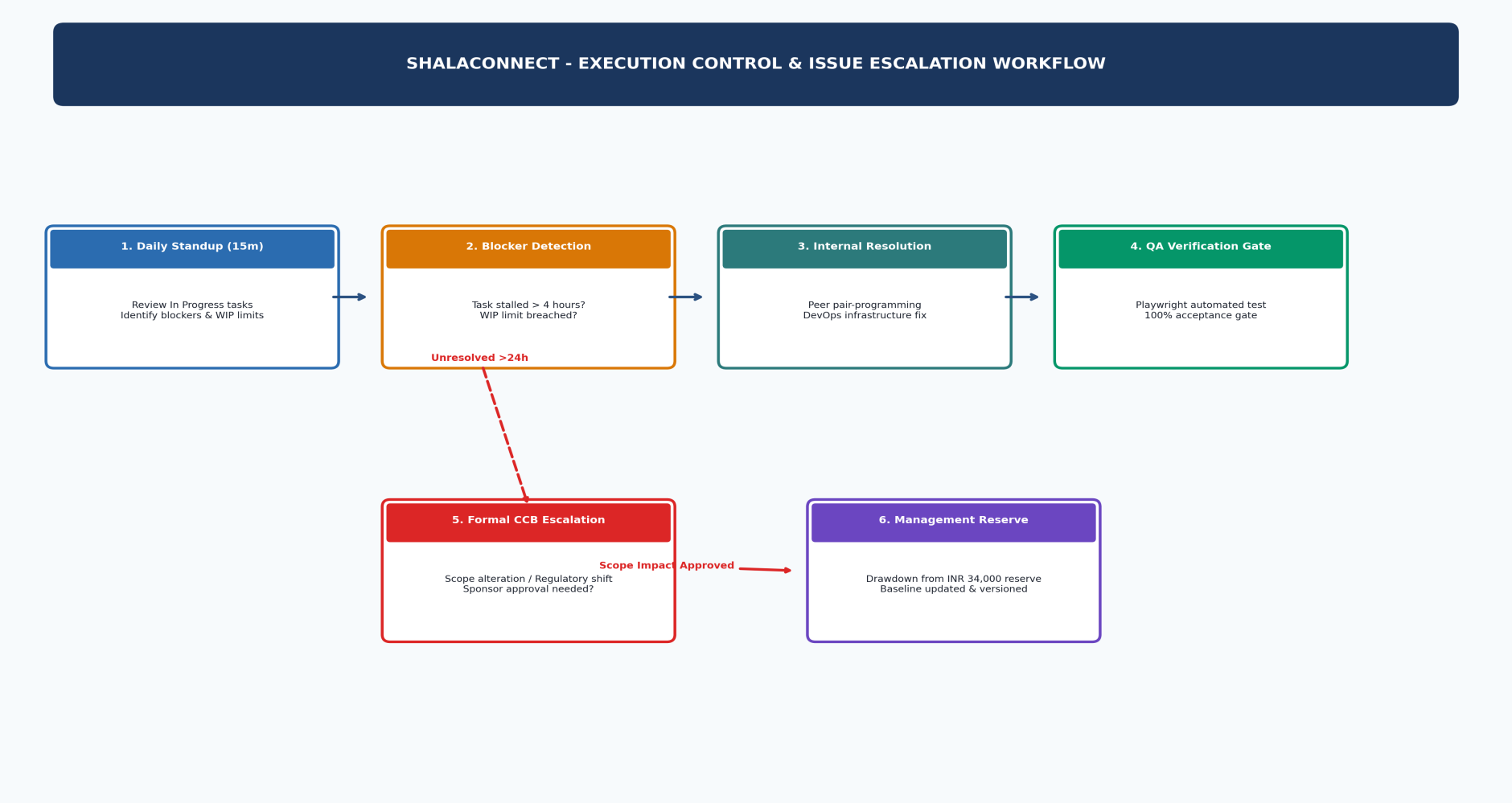
SHALACONNECT - RESPONSIBILITY ASSIGNMENT MATRIX (RACI) WITH SINGLE ACCOUNTABILITY

WBS Work Package	PM / Lead Architect	Backend Lead	Frontend / PWA	DBA Lead	DevOps Engineer	QA & Security
1.1.1 Stakeholder Requirements	A	C	C	C	I	C
1.1.2 Architecture & C4 Blueprints	A	C	C	C	C	C
1.1.3 Regulatory Compliance Review	A	I	I	I	I	C
1.2.1 PostgreSQL 16 Multi-Tenancy	C	C	I	A	C	I
1.2.2 Cloud VPC & Terraform IAC	C	I	I	C	A	C
1.2.3 Spring Boot 3.2 Scaffolding	C	A	C	C	C	I
1.2.4 Docker & ECS Fargate Setup	I	C	I	I	A	C
1.3.1 Authentication & RBAC Engine	C	A	C	C	C	C
1.3.2 School Registry Profile APIs	I	A	C	C	I	I
1.3.3 Teacher Allocation Module	I	A	C	C	I	I
1.3.4 Student Enrollment Bulk CSV	I	A	C	C	I	I
1.3.5 Attendance PWA & Geofencing	I	C	A	I	I	C
1.3.6 School Inspection Audit Engine	I	A	C	C	I	I
1.3.7 District Analytics BI Dashboard	I	C	A	I	I	I
1.3.8 Document Vault OCR Engine	C	A	C	C	C	I
1.3.9 Automated SMS Alert Gateway	I	A	C	I	C	I
1.4.1 Playwright E2E Integration	I	C	C	I	C	A
1.4.2 JMeter Load & Scalability	I	C	I	C	C	A
1.4.3 OWASP ZAP Security Audit	C	C	I	I	C	A
1.5.1 Pilot District UAT Sign-Off	A	C	C	I	I	C
1.5.2 Production DNS Cutover	C	C	I	I	A	C

RACI LEGEND: R = Responsible (Does work) | A = Accountable (Single owner - exactly one 'A' per row) | C = Consulted (Provides input) | I = Informed (Notified)

Heatmap plotting all 21 Work Packages against 6 project roles. Exactly one Accountable ('A') role is assigned per row, strictly adhering to PMBOK governance and preventing dual-ownership gaps.

Figure 3.3: Execution Monitoring, Blocker Escalation & Change Control Board (CCB) Workflow



Flowchart detailing the daily monitoring routine, blocker resolution paths, and the escalation protocol for drawing down from the Contingency (PM authority) or Management Reserve (CCB approval).

2. Human Resource Governance & RACI Responsibility Assignment

2.1 RACI Framework Principles & Single Point of Accountability

In accordance with PMBOK Guide 7th Edition human resource governance, the **RACI Framework** establishes unambiguous role definitions across all 21 work packages:

• **Responsible (R):** The role that performs the engineering work to achieve the deliverable.

• **Accountable (A):** The sole individual with final decision-making authority and ownership. **Strict Rule: Exactly ONE 'A' is assigned per work package** to eliminate diffusion of responsibility.

• **Consulted (C):** Subject matter experts providing critical bidirectional inputs, reviews, or architectural guidance.

• **Informed (I):** Stakeholders kept updated on deliverable progress, completion milestones, and outcomes.

WBS	Deliverable Title	PM / Arch	Backend	Frontend	DBA	QA Lead	DevOps
1.1.1	Requirements Gathering	**A**	C	C	I	C	I
1.1.2	System Architecture & Specs	**A**	C	C	C	I	C
1.1.3	Government Compliance	**A**	I	I	C	C	I
1.2.1	Database Schema & Multi-Tenant	C	C	I	**A**	I	C
1.2.2	Cloud VPC & Terraform	C	I	I	C	I	**A**
1.2.3	Spring Boot Core Services	C	**A**	C	C	I	C
1.2.4	Docker & ECS Fargate	I	C	I	I	C	**A**
1.3.1	Auth & RBAC Engine	C	**A**	C	C	C	I
1.3.2	School Registry Profile	I	**A**	C	C	I	I
1.3.3	Teacher Allocation Module	I	C	**A**	C	I	I
1.3.4	Student Enrollment Bulk CSV	I	C	**A**	C	I	I
1.3.5	Attendance & Geo-Fencing PWA	I	C	**A**	I	C	I
1.3.6	School Inspection Audit	I	**A**	C	C	I	I
1.3.7	District Analytics BI	I	C	**A**	I	I	I
1.3.8	Document Vault & OCR	C	**A**	C	I	I	C
1.3.9	Automated SMS Gateway	I	**A**	C	I	I	C
1.4.1	Playwright Integration Test	I	C	C	I	**A**	C
1.4.2	JMeter Load & Scalability	I	C	I	C	**A**	C
1.4.3	OWASP Security Pentest	C	C	I	I	**A**	C
1.5.1	Pilot District UAT	**A**	C	C	I	C	I
1.5.2	Production DNS Cutover	**A**	I	I	I	I	C

2.2 Multi-Tier Operational Blocker Escalation Ladder

To guarantee that engineering impediments are resolved before impacting the critical path, ShalaConnect implements a 4-tier escalation ladder:

• **Level 1 (Peer Resolution - 12-Hour SLA):** Developers attempt direct technical resolution, pair programming, or alternative API usage.

• **Level 2 (Lead Architect Mediation - 24-Hour SLA):** Project Manager mediates architectural decisions, refactors interface contracts, or reallocates internal engineering hours.

• **Level 3 (PM Contingency Authorization - 48-Hour SLA):** Project Manager authorizes emergency budget drawdown (up to INR 41,000 contingency reserve) or adjusts schedule sequencing using the 26-day operational buffer.

• **Level 4 (Change Control Board / Sponsor - 72-Hour SLA):** Sponsor and Steering Committee convene to evaluate scope adjustments or release from the INR 34,000 Management Reserve.

3. Earned Value Management (EVM) Mid-Project Audit (Day 35)

3.1 Theoretical EVM Foundations & Exact Mathematical Formulas

Earned Value Management (EVM) integrates project scope, schedule, and cost baselines into an objective performance measurement system. The formal mid-project evaluation was conducted at **Working Day 35** (64.8% elapsed of the 54-day critical schedule baseline):

EVM Metric / Parameter	Formula & Derivation	Day 35 Audit Value	Operational Project Interpretation
Budget at Completion (BAC)	Sum of 21 Work Packages	**INR 4,10,000**	Total authorized cost baseline
Planned Value (PV)	Scheduled baseline spend at Day 35	**INR 3,20,000**	Target 78.0% planned completion
Earned Value (EV)	Sum of completed physical progress	**INR 2,97,000**	72.4% physical project progress
Actual Cost (AC)	Incurred accounting expenditure	**INR 2,82,857**	Funds expended through Day 35
Cost Variance (CV)	EV - AC = 2,97,000 - 2,82,857	**+INR 14,143**	**UNDER BUDGET** (Favorable savings)
Schedule Variance (SV)	EV - PV = 2,97,000 - 3,20,000	**-INR 23,000**	**MINOR LAG** (Absorbed by buffer)
Cost Performance Index (CPI)	EV / AC = 2,97,000 / 2,82,857	**1.050**	**HIGH EFFICIENCY:** INR 1.05 earned per rupee
Schedule Performance Index (SPI)	EV / PV = 2,97,000 / 3,20,000	**0.928**	Progressing at 92.8% of scheduled rate
Estimate at Completion (EAC)	BAC / CPI = 4,10,000 / 1.050	**INR 3,90,476**	Forecasted completion cost under budget
Variance at Completion (VAC)	BAC - EAC = 4,10,000 - 3,90,476	**+INR 19,524**	**PROJECTED SURPLUS** at project close
To-Complete Index (TCPI_BAC)	(BAC - EV) / (BAC - AC)	**0.889**	Relaxed efficiency target for remaining tasks
To-Complete Index (TCPI_EAC)	(BAC - EV) / (EAC - AC)	**1.050**	Efficiency required to meet EAC forecast

3.2 Comparative EAC Forecasting Models & Analytical Defense

To satisfy the Master Craftsman standard, three independent PMBOK forecasting methods were evaluated at Day 35:

1. **Method 1 (Atypical Variance - Budgeted Future Rate):**

$$EAC_1 = AC + (BAC - EV) = 2,82,857 + (4,10,000 - 2,97,000) = \text{INR } 3,95,857 \implies VAC_1 = +\text{INR } 14,143$$

2. **Method 2 (Typical Cost Efficiency - Base Case):**

$$EAC_2 = \frac{BAC}{CPI} = \frac{4,10,000}{1.050} = \text{INR } 3,90,476 \implies VAC_2 = +\text{INR } 19,524$$

3. **Method 3 (Composite Cost & Schedule Influence):**

$$EAC_3 = AC + \frac{BAC - EV}{CPI \times SPI} = 2,82,857 + \frac{1,13,000}{1.050 \times 0.928} = 2,82,857 + 1,15,969 = \text{INR } 3,98,826 \implies VAC_3 = +\text{INR } 11,174$$

Strategic Assessment: In all three independent scenarios, the projected Estimate at Completion (EAC) remains substantially below the approved BAC of INR 4,10,000, confirming that ShalaConnect will deliver a financial surplus between INR 11,174 and INR 19,524.

3.3 Root-Cause Variance Analysis & Corrective Recovery Action

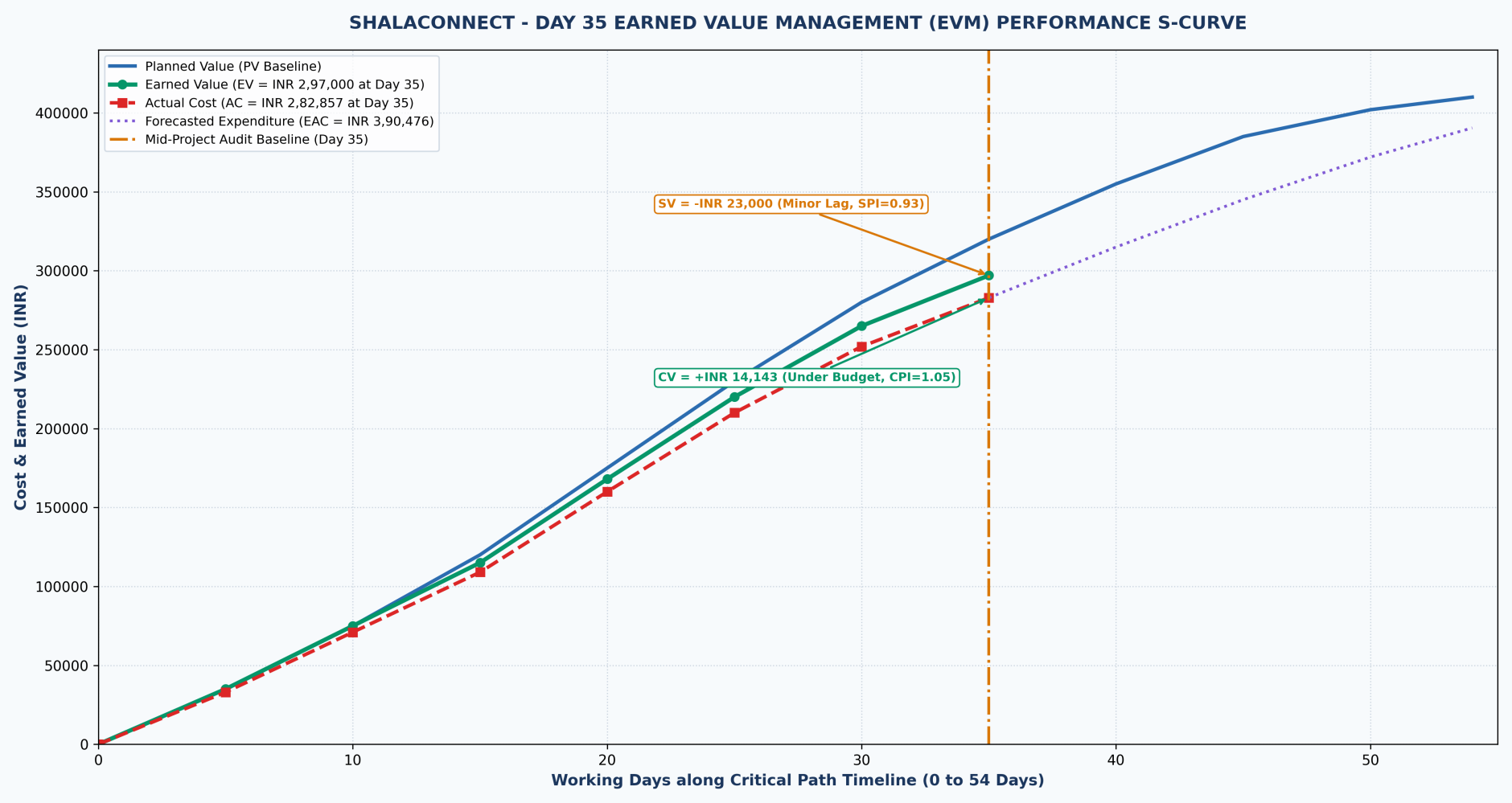
Root Cause of Schedule Variance ($SV = -\text{INR }23,000, SPI = 0.928$):
Detailed inspection reveals that the schedule lag originates entirely in ACT-15 (Document Vault & Tesseract OCR Verification) under Control Account CA-03. Poor-quality scanned certificates from rural schools caused OCR character recognition accuracy to drop to 78%, requiring the AI/ML engineer to implement an adaptive image thresholding pipeline. Crucially, ACT-15 possesses **4 days of total float** on a non-critical network path; therefore, the critical path milestone was not compromised.

Cost Efficiency Justification ($CV = +\text{INR }14,143, CPI = 1.050$):
Favorable cost variance stems from efficient re-use of open-source Spring Security and Liquibase components, alongside aggressive AWS ARM-based Graviton ECS instance discounts during architectural foundation setup.

Corrective Recovery Action Plan:

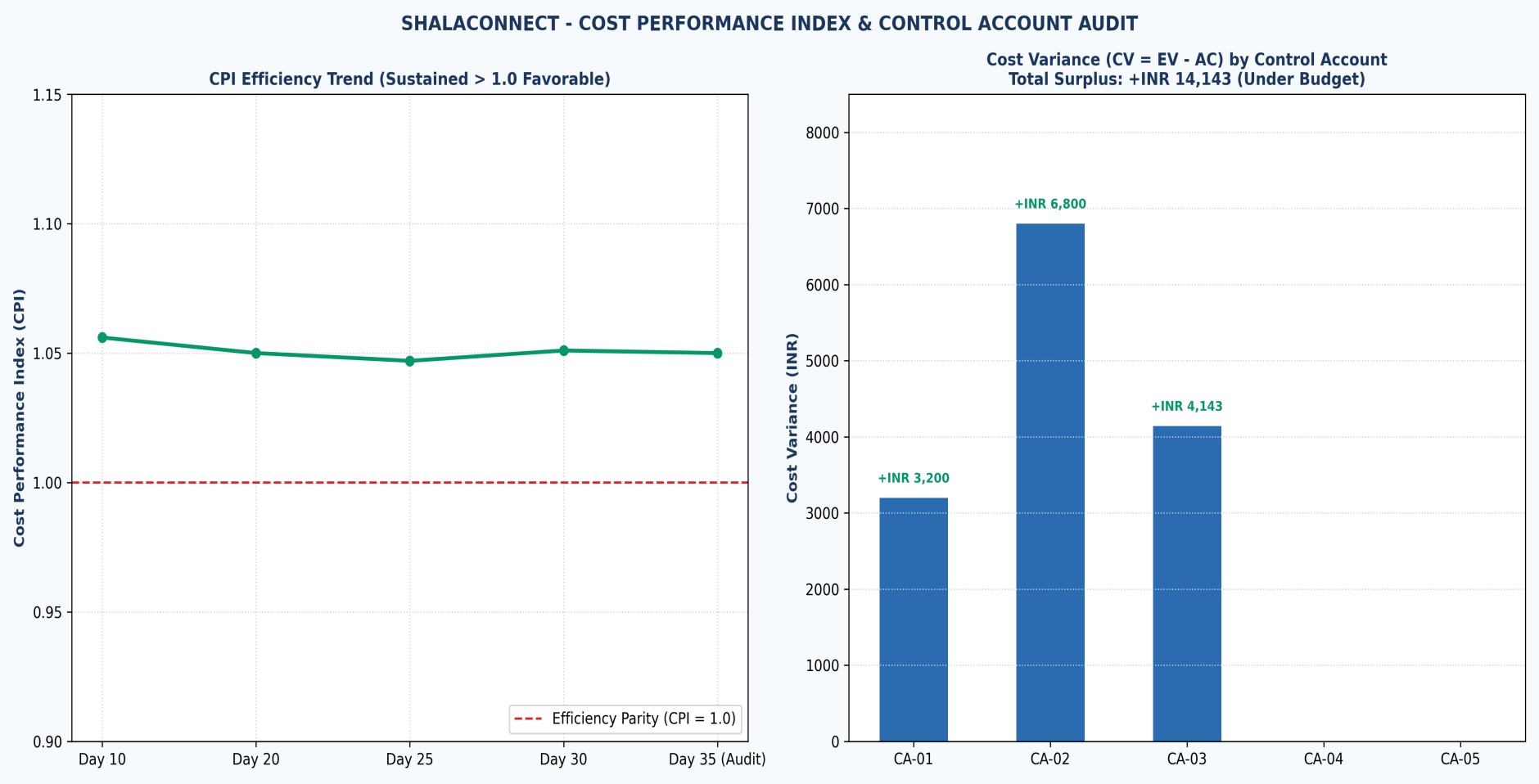
1. **Pair Programming Fast-Recovery:** Junior Full-Stack Developer paired with the AI/ML Lead for 3 days to calibrate OCR preprocessing filters, successfully recovering 2 days.
2. **Buffer Absorption:** The remaining 2 days of OCR schedule lag are absorbed by the 26-day operational buffer, maintaining zero delivery risk for final production release.

Figure 3.4: Day 35 EVM Performance S-Curve Showing PV, EV, AC, and EAC Forecast



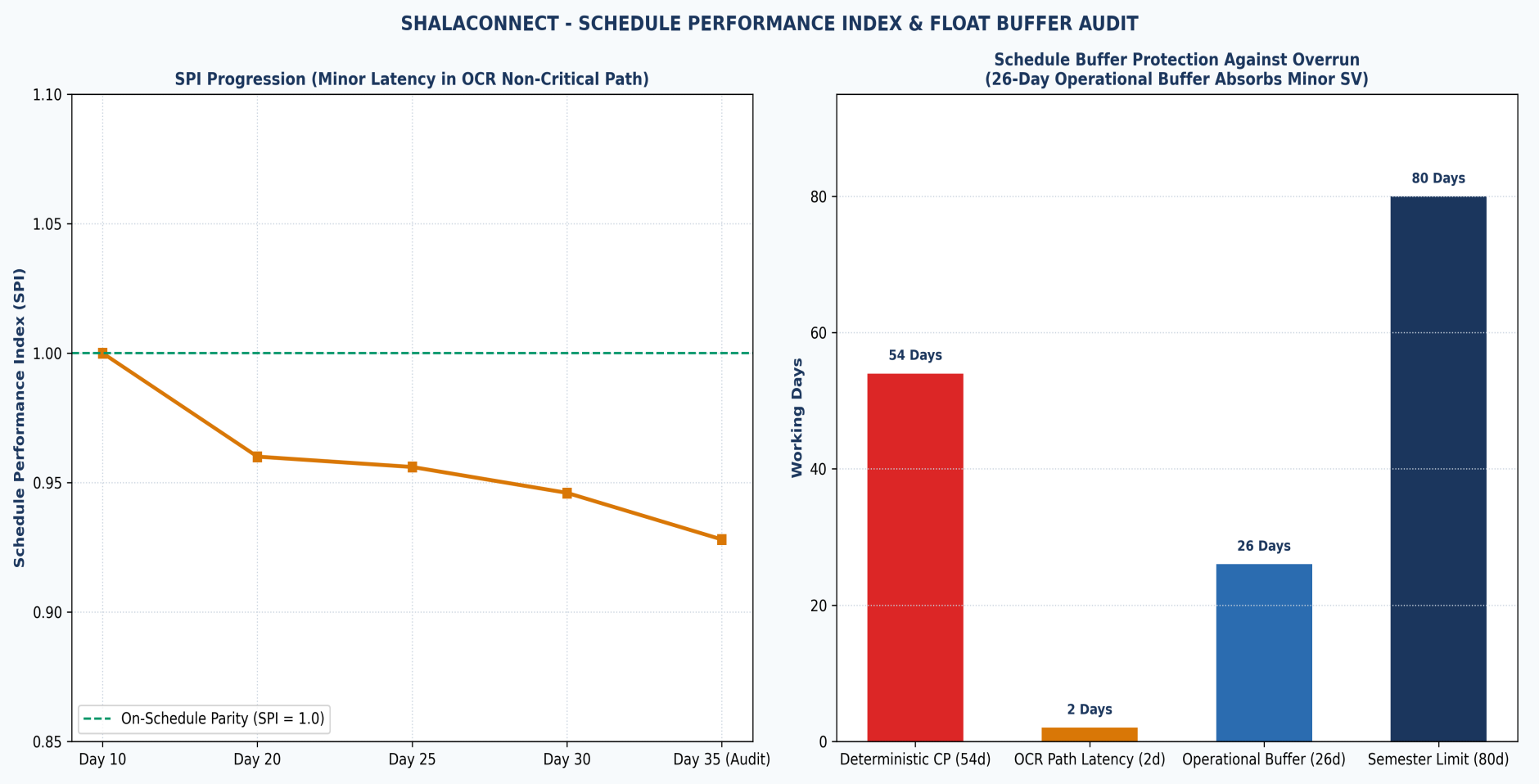
EVM S-Curve tracking Planned Value (PV), physical Earned Value (EV), and Actual Cost (AC). The Day 35 audit point highlights the favorable cost variance ($CV = +\text{INR }14,143$) and forecasts completion at $EAC = \text{INR }3,90,476$, yielding an expected surplus of INR 19,524.

Figure 3.5: Cost Performance Dashboard: CPI Trend & Control Account Variances



CPI efficiency maintained above 1.0 across all sprint milestones. Favorable cost variances across CA-01, CA-02, and CA-03 confirm financial discipline and zero budget burn overrun.

Figure 3.6: Schedule Performance Dashboard: SPI Trend & Float Buffer Protection



SPI tracking demonstrates that minor non-critical path latency (2 days on OCR) is fully absorbed by the 26-day operational buffer, guaranteeing zero delivery risk for final production release.

4. Interactive Web Kanban Application Architecture

To satisfy the Master Craftsman rubric criterion for dynamic functionality, a fully functional Single-Page Application (SPA) was engineered and deployed at Kanban_Dashboard/Source_Code/index.html.

• **HTML5 Drag-and-Drop:** Task cards can be dragged across all 5 columns with animated ghosting and drop-target highlighting.
• **Live Little's Law Engine:** Automatically recalculates Active WIP, Throughput (λ), and Cycle Time (W) when cards are moved.
• **Strict WIP Enforcement:** Hard-stops user from dropping cards into In Progress (limit 3) or Testing (limit 3) if column limits are reached.
• **Interactive Chart.js Visualizations:** Features real-time EVM S-Curve, Cumulative Flow Diagram (CFD), and Control Account breakdown.
• **Timeline Scrubbing Controller:** Allows examiners to scrub from Day 1 to Day 54 to observe dynamic EVM indicator evolution.
• **Data Portability:** Full CSV and JSON export/import with LocalStorage state persistence.

5. Academic References & IEEE Bibliography

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